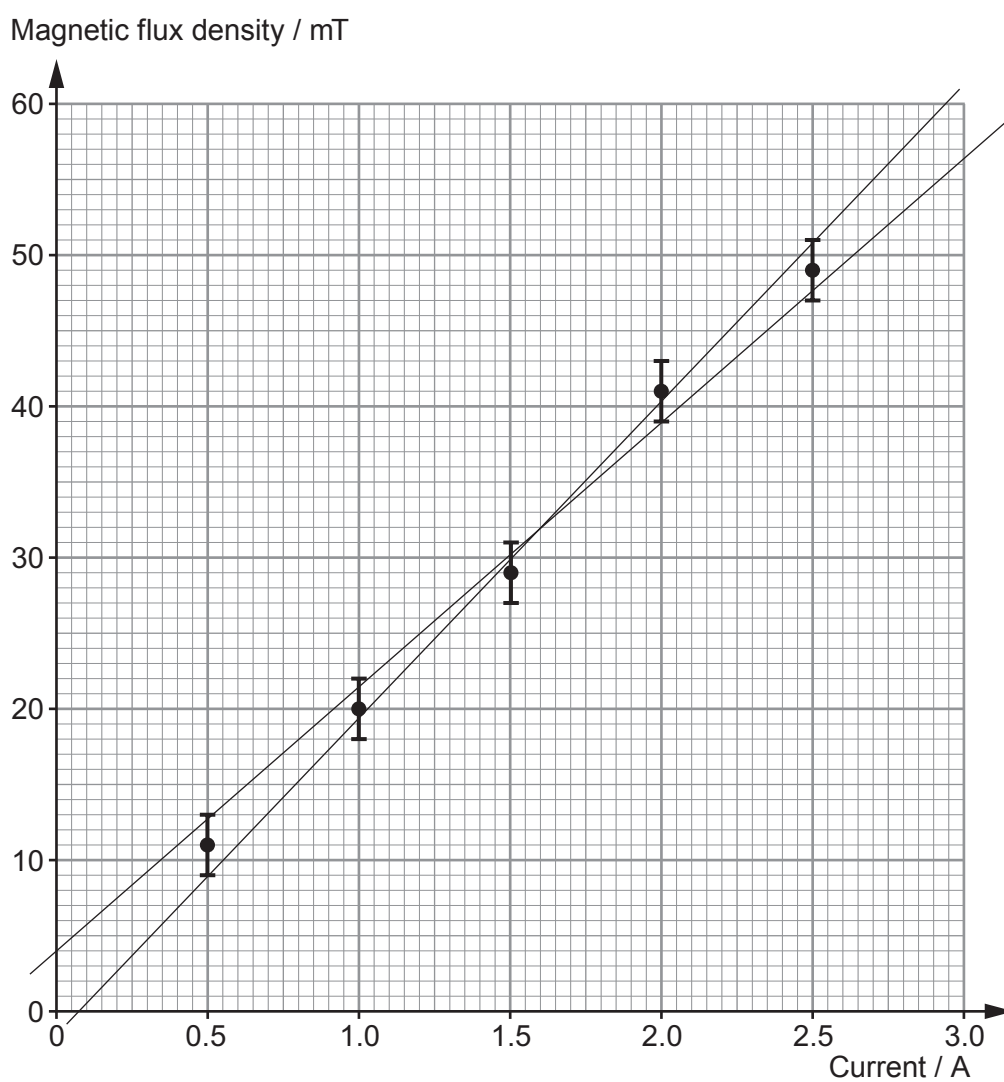




- (a) Calculate the mean value of the gradient along with its **percentage** uncertainty. [4]

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- (b) Calculate the number of turns in the solenoid along with its **absolute** uncertainty. The percentage uncertainty in the length of the solenoid is 0.4%. [3]

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- (c) (i) The solenoid manufacturer states that there are exactly 5000 turns in the solenoid. Evaluate the accuracy of your value obtained in part (b) and whether or not the graph is in agreement with the equation: $B = \mu_0 nI$. [4]

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- (ii) Suggest a reason for the disagreement between the manufacturer's stated value (5000 turns) and your value calculated in part (b). Suggest how the experimental technique might be improved for better agreement. [2]

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- (iii) A metal plate of thickness 1.4 mm is used to reduce the intensity of the X-rays produced to 60 % of the incident intensity. If a second identical plate is now also placed in the beam, calculate the new transmitted **percentage** intensity. [3]

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- (b) (i) Describe how the Doppler shift principle can be used to measure the speed of blood through an artery. [2]

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- (ii) Ultrasound of frequency 2 MHz was used to calculate the speed of blood and a Doppler shift of 200 Hz was detected. The measurement was taken at an angle of 40° to the direction of flow and the speed of ultrasound through the blood is 1500 m s^{-1} . Calculate the speed of blood flow. [2]

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- (c) (i) Describe the properties of technetium-99m (Tc-99m) that make it such a good radioisotope in the effective diagnosis of medical problems. Justify your choice of properties. [3]

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- (ii) Explain clearly how a gamma camera is used to detect the gamma rays given off by a technetium-99m source. [3]

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- (iii) In positron emission tomography a positron annihilates an electron producing two photons of energy 0.511 MeV. By setting out your reasoning clearly, determine whether or not the value of the photon energy is correct. [3]

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